

Adaptive Learning, Diagnostic Assessment and Personalized Path-Planning Engine

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1. Background

Most online courses show lessons in one fixed order for everyone. In reality, learners are very different. Some have never seen the topic, some already know half of it, and their roles, deadlines and free time also vary. A fixed sequence cannot fit both a beginner who has only 10 minutes a day and an experienced person who just wants a short refresher. The first learner gets overwhelmed because lessons are too hard or too long, and the second wastes time on things they already know.

The idea behind this project is to replace this static engine with a personalization engine. It should find out what a learner already knows, plan what to study within their goal, deadline and daily time limit, and keep adjusting as they progress, while always respecting the order in which topics depend on each other. It should also explain why each lesson was chosen, so the learner can trust the plan.

2. Problem Statement

The problem is to design and build a personalization engine that, for a given learner, does the following:

- Estimates the learner's initial and ongoing skill mastery.
- Stores the curriculum prerequisites, so the order of topics is always known.
- Generates a day-by-day study plan within the learner's goal, deadline and daily time budget (5, 10, 15 or 30 minutes).
- Adapts the difficulty of the content using an explicit rule.
- Schedules remediation and spaced review.
- Replans when the learner misses days, fails an assessment or changes the deadline or budget.

Every decision must be explainable, and mandatory prerequisites must never be broken. The scope is the engine, its evaluation and its explanations. Building a full learning platform, authoring real course content or using private learner data is outside the scope.

Key challenges I have identified:

- The initial skill level has to be estimated from a very short diagnostic, and the estimate must carry uncertainty instead of being a single number.

- Prerequisites are strict rules, whereas time, deadline and difficulty are things to optimize. If the goal cannot fit in the available time, the system should say so clearly and not silently break a rule.
- New lessons, remediation and spaced review all compete for the same few minutes each day, so a clear priority policy is needed.
- After missed work or changed constraints, the new plan must be valid and keep everything already completed.
- Each inclusion, omission, review or difficulty change must be traceable to learner state, graph constraints or a policy score.

3. Expected Output

The project will deliver a working, documented prototype and an evaluation report. The main outputs are::

- Curriculum and learner-state model: a queryable graph of concepts, lessons, exercises, durations and prerequisites, with a mastery estimate and uncertainty for every skill.
- Adaptive diagnostic: a short, bounded test that gives a calibrated starting skill estimate, compared against fixed and random question selection.
- Time-budgeted path generator: a prerequisite-valid, day-by-day plan for a given goal, deadline and daily budget.
- Online mastery update and difficulty policy: mastery is updated after each quiz or exercise, and the next difficulty follows an explicit policy.
- Spaced review and remediation: reviews scheduled by forgetting risk, prerequisite importance and remaining daily time.
- Dynamic replanning: a valid revised plan after missed days, failed assessments, or changed deadlines and budgets.
- Explainability: a reasoning attached to every lesson, omission, review and difficulty change.
- Comparative evaluation: results for diagnostic accuracy, plan feasibility, runtime, mastery prediction and calibration, and simulated learning gains, compared with static and greedy baselines.

By the end of the project, the engine should be able to give different valid plans to several synthetic learners who differ in skills, deadlines and daily time budgets. These plans should update dynamically when the learner shows new evidence of progress or misses work, and every decision should come with a clear explanation. The engine's performance will be measured against non-adaptive baselines, and the aim is to show a measurable improvement over them.